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Females Welcome?

Gender-specific immigration and natives' subjective well-being

Abstract

Does the gender of immigrants affect the subjective well-being of natives? Our analysis is based on representative data from Germany and Catalonia (Spain). Both regions represent two most-different EU cases in terms of macroeconomic context, political mobilisation, and typology of migrants. Fixed-effects panel models and a regression discontinuity analysis provide robust evidence that the arrival of migrants is positively associated with the happiness of natives in general. Yet, when we identify crisis years and decompose the sample between male and female immigrants, we find significant differences. While female immigrants are perceived as a source of happiness (positive relationship), male immigrants are often perceived as a threat (negative relationship). Our findings have important implications for the understanding of anti-immigration behaviours and possible political responses.

Keywords Causality, Germany, immigration, gender, sex, Spain, subjective well-being

Word count 7767 (incl. references)

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Introduction

How does the immigration of men and women affect the subjective well-being (SWB) of the native population in host countries? Recently, western media covers scenes of a warm welcome to war refugees from Ukraine as it also did to fleeing Syrians in 2015. Still, it is often assumed that migrants have a negative influence on the people in the host country. Right-wing populist parties routinely blame immigration for cultural, economic, and social problems in their societies. The perception of immigration as a source of trouble is also widespread among native populations exposed to refugees (Dinas et al., 2010; Karakayali, 2018) or to economic migrants from Muslim countries (Strabac et al., 2014).

While the prevalence and severity of anti-immigration sentiments is obvious, the relation to the subjective well-being of the citizens expressing them is not. A growing number of studies have focused on the life satisfaction of migrants (e.g. see: Helliwell et al., 2020; Hendriks, 2015; Angelini et al., 2015) and their motivations to migrate compared to stayers (Polavieja et al., 2018). Yet, there are only a few studies addressing how the influx of immigrants is related to the subjective well-being of native populations (e.g. see: Grover and Helliwell, 2019; Ivlevs and Veliziotis, 2018, O'Connor 2020). This oversight is unfortunate because a better understanding of the link between immigration and the happiness of native populations can help to gauge when and why immigration sparks strong politicisation (Colantone and Stanig, 2018). It can also enlighten the still unsettled questions of whether, and under what conditions, ethnic heterogeneity is harmful to social cohesion (Putnam, 2007; Van der Meer and Tolsma, 2014).

Our hypotheses derive from three theoretical perspectives on migration. According to the *social production function theory* (Ormel et al., 1999), an increasing number of newcomers should positively affect the subjective well-being of the domestic population on average, as immigrants add to local productive resources, consumption, and economic growth. However, during times of macroeconomic crisis, an influx of immigrants may also increase perceived ethnic rivalry for scarce

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resources, create status anxiety and uncertainty which should make natives unhappy (*ethnic competition theory;* Olzak, 1992).

What is more, the gender of newcomers can moderate the relationship between sheer immigration numbers and locals' subjective well-being. Normative *expectation states theory* (Correll and Ridgeway, 2006) and related empirical studies confirm deep-rooted gendered norms and behaviour. Thus, we expect that, on average, the arrival of female migrants is anticipated to be a gain in social welfare and a plus for the subjective well-being of the locals. While male migrants may be more often perceived as a threat to a tight labour market, austere welfare state, or public safety, and declines the subjective well-being of the host population.

Germany is our primary test case. It is the second most popular destination of international migration in the world (International Organization of Migration (IOM), 2020). German findings will be cross-validated with a most-different EU case in terms of macroeconomic context, political mobilisation and typology of migrants: Spain. If the hypotheses hold in the diverse contexts of Germany and Spain and with different model strategies, we expect them to be robust in other contexts and relevant for policymakers.

The rest of the paper is organised as follows. The next section reviews the theoretical and empirical literature on the relationship between immigration, gender, and natives' subjective well-being. Subsequently, we explain the case selection, describe the data, and statistical methods. Next, we present our findings. The final section concludes.

Background

Immigration and the Happiness of the Native Population

Subjective well-being is commonly defined as the self-reported cognitive and emotional appraisal of one's life (Diener et al., 2002). It is broadly equated with happiness in the social science literature (Easterlin, 2004; Veenhoven, 2012). Research on immigration and happiness has often compared the subjective well-being of migrants with that of the native population of the host and

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origin country. While this literature shows that migrants report lower subjective well-being than the native host population (e.g. Bălţătescu, 2007; Bartram, 2011; Sand and Gruber, 2018), conflicting findings appear when migrants are compared to native stayers in the origin country (e.g. Cuellar et al. 2004; Vanhoutte et al., 2019; Hendriks, 2015; Melzer, 2011; Frijters et al. 2004; Amit and Riss, 2014).

While all this research sheds light on important aspects of the relationship between immigration and subjective well-being, very few empirical studies have shown how the influx of immigrants affects the happiness of natives. Those studies show either a slightly positive (Akay et al. 2014; Betz and Simpson, 2013), slightly negative (Howley et al., 2020) or an insignificant relationship (Helliwell et al., 2018; Ivlevs and Veliziotis, 2018; O'Connor, 2020).

To systematically disaggregate these incoherent findings, we make use of three theoretical perspectives. First, a positive relationship between immigration and natives' happiness can be explained by *the social production function theory* (Ormel et al., 1999). Increasing numbers of newcomers should lift the overall subjective well-being of the domestic population, as immigrants generally grow the national labour force and add to the country's productivity. This is even more important for aging societies like Germany, where for decades fertility is significantly below the net-reproduction rate ([reference excluded to ensure anonymity]). Beyond materialistic utility, positive net-migration may also benefit the local population by a revelation effect that reminds them of living in an attractive, appreciated, and eminently respected place (Kourtit et al., 2022).

Most empirical studies in economics of international migration confirm positive short and long-term effects of migration on economic growth (Alesina et al., 2016; D'Ambrosio et al., 2019; Rodríguez-Pose and Von Berlepsch, 2014, 2015). Oftentimes immigrants tend to work in precarious, low-skilled, unstable and poorly paid jobs that natives do not want as they lack crucial resources like legal status, economic means, cultural expertise, or social capital to compete with natives (Venturini and Villosio, 2018; Lewis et al., 2015). The inferior and insecure status of newcomers can also trigger a warm-glow feeling among natives of being respected, helpful and giving (Andreoni, 1990; Gürek and Kasulke, 2021).

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Importantly, this overall effect does not imply that each native citizen has to directly interact with newcomers in order to experience happiness. Rather, media coverage, virtual encounter on social media platforms, or hearsay may suffice to increase subjective well-being (e.g. see: Anon, 2022; Fischhaber, 2015). Thus, along this line of reasoning and evidence, we expect that, on average, rising numbers of newcomers increase the subjective well-being of the native population (*Hypothesis 1*).

Still, further differentiation is important as countries experience economic up- and downturns, times of political instability, and other external crises. The impact of immigration on local wages and unemployment (Card, 2005; Ottaviano and Peri, 2012), two core determinants of happiness, are mixed. When adverse shocks hit, people begin to fear a loss of control, social status, jobs, and life as they know it with negative repercussion on happiness (Clark, 2010; Di Tella et al., 2001; Ferrer-i-Carbonell and Ramos, 2014). The arrival of newcomers may then be perceived as a social threat (Helliwell et al., 2020), provoking problematic social comparisons and a perceived rivalry for jobs and welfare benefits.

Ethnic competition theory (Olzak, 1992) and related empirical studies (Becker et al., 2017; Laurence and Bentley, 2018; Rooduijn and Burgoon, 2018; Van der Zwan et al., 2017) help to predict negative effects on the subjective well-being of natives when they are insecure and feel they have limited resources at their disposal. In crisis times, rising numbers of immigrants attract public attention, trigger comparisons, and likely increase perceived competition. This dynamic can be stronger when immigrants who do not share the same kind of ethnic, cultural, and religious background as they may be less well understood (Newman, 2013). For instance, the 2008 Global Financial Crisis correlates with less positive attitudes towards immigrants in Europe (Vogt Isaksen, 2019) and mobilized an anti-immigration opposition in the US (Creighton et al., 2015). Similarly, during the so-called 'migration crisis' in 2015-17, Stockemer et al. (2020) discover a solid relationship between more critical attitudes toward immigrants in the European Social Survey. Therefore, we expect that crisis events rub off on the perception of immigrants. Accordingly, external "shocks" could negatively impact the subjective-well-being of natives (*Hypothesis 2*).

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The Role of Gender

Research on the effect of the gender of the immigrants on the subjective well-being of natives is missing. Arguably, however, gender is an important factor. *Expectations states theory* in sociology (Correll and Ridgeway, 2006) claims that societies nurture different performance expectations depending on individuals' social categories, and that these expectations reinforce systems of social status hierarchies. Gender status beliefs have traditionally made men appear more competent, capable, and valuable than women (Ridgeway, 2001). Men are also perceived to be "better at some particular tasks (e.g. mechanical tasks), while women are better at others (e.g. nurturing tasks)" (Correll and Ridgeway, 2006:32). Partly these differences in perception reflect the traditional gender roles of male "providers" and female "care takers" (Becker, 1985). As a result, gender inequality continues to be reproduced in all societies (Ridgeway and Bourg, 2004).

Substantial empirical evidence confirms the importance of gender status beliefs. For instance, experiments in social psychology show that men are more likely to embrace competition and women more likely to shy away and underperform at the same level of ability (e.g. Niederle and Vesterlund, 2011; Reuben et al., 2020). In a similar vein, recent economic and sociology experiments show that gender status beliefs influence credit attribution for group work (Sarsons et al., 2021). When task performance results in social status rankings, men try and correctly solve more tasks than women (Schram et al., 2019).

Thus, we expect gender roles or gender status beliefs to shape natives' perceptions of newcomers. If men are seen as more capable, competitive, and aggressive than women, and women as more social, "caring", and compliant than men, male immigrants will be more often perceived as a threat to local workers, employees, and to the public, while female immigrants might be seen as a gain for the formal labour and informal care market, as well as for the society as a whole (Shah and Lerche, 2020). Indeed, Venturini and Villosio (2018) show a strong gender segmentation on migrant labour markets with women focusing on homecare and healthcare sectors, and men mainly employed in manufacturing and construction. Stereotypical gender perceptions should also hold in other social

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contexts, such as education, sports, politics, marriage markets, or crime. For instance, the unconfirmed but widely believed relationship between immigration status and crime rate (Boateng et al., 2021) may particularly affect male immigrants as men of all countries and at all times show more criminal behaviour than women (e.g. Eddy, 2016). Against this background, we hypothesize that local populations perceive the influx of female migrants more positively (as a resource) and the influx of male migrants more negatively (as a threat). The former should be associated with an increase, the latter with a decrease of subjective well-being (*Hypothesis 3*).

Case selection: Germany and Catalonia

Our analysis focuses on Germany - a highly relevant immigration country and a well-documented case. We control for within-country heterogeneity between East and West Germany (Auspurg et al., 2016), and we add a second rich panel data source to replicate and cross-validate the findings. The case selection of regional Spanish data follows a most-different design logic within the EU. It helps to ensure robust and ecological valid results (Seawright and Gerring, 2008). Within the EU context, Germany and Spain experienced different inflows of migrants, and underwent different macroeconomic and political developments during the last decades.

Germany is the second-largest destination for migrants worldwide. The country has experienced an increase in the foreign-born population 'from 8.9 million in 2000 to 13.1 million in 2019' (IOM, 2020:25). In 2018, more than a quarter of the German population (25.5%) had a migration background. This includes all persons who themselves or whose mother or father did not have a German passport (Bundesamt für Migration und Flüchtlinge (BAMF), 2020b). (West) Germany's immigration waves have changed in composition. Immediately after WWII, Germany absorbed millions of displaced Germans. During the 1950s until the oil crisis in 1973, the country hired 14 million economic migrants. Many of these 'guestworkers' remained and brought their families to Germany particularly during the 1980s. After the end of the Cold War, asylum seekers and German 'resettlers' from Eastern Europe came to Germany. Asylum applications rose more than fourfold, to 440,000 in

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1992. While immigration has intermittently fallen, numbers began to rise again in 2010. In 2015, Germany experienced an unprecedented influx of refugees mostly from Syria. Net migration reached 1,139,402 persons that year (BAMF, 2020b). In our analysis, we particularly focus on this event.

Germany is also the largest EU economy. It managed the 2008 Financial Crisis with a mix of passive, deficit-spending unemployment aid, short-time work arrangements, and active economic stimulus packages. In contrast to other EU member states, unemployment actually fell from 11.2% (2005) to 8.7% in 2007, 5.4% in 2012, and 3.1% in 2019 (OECD, 2022).

For a long time, Germany did not have strong right-wing populist parties (Berbuir et al., 2015). However, with the founding of the Alternative für Deutschland (AfD) in 2013, right-wing anti-migration views became vocal in electoral politics (Wurthmann et al., 2021).

The situation in Spain is different. Since the early 2000s, Spain has been the host country for economic migrants from Latin American, European, and African countries. Many of these immigrants moved to one of the wealthiest regions of Spain, Catalonia, where migrants currently constitute 14% of the population (Spanish Statistical Office, 2021b). Catalonia, like Spain in general, suffered from a severe economic downturn after the Financial Crisis in 2008 (OECD, 2022). It registered with the highest growth in inequality and unemployment in Europe (OECD, 2016). From 2007 to 2012, the unemployment rate rose from 8.6% to 25.8% (Spanish Statistical Office, 2021b). Also, in contrast to Germany, strong polarisation emerged on both sides of the political spectrum in Spain.

Materials and Methods

Our data stem from German and Catalan individual-level panel surveys. Migration data is from national statistical offices.

Panel Data

The German Socio-Economic Panel (GSOEP), an annual panel survey of a representative German sample of households, is a well-established source for well-being research (Goebel et al.,

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2019). We use all waves from 1990 to 2017 except for the innovation, birth cohorts, and family type subsamples. The uploaded data consist of 56,974 people, of whom 10,415 are first-generation immigrants (migrated from 1950 and later, 1,022 migrated earlier), 3,474 (2nd generation migrations) have an indirect migration background, and 42,063 have no migration background. The average length of stay was 9.09 years. The final dataset consisted of 507,188 person-years. Missing data were deleted in a pair-wise manner.

Catalonia hosts the only available longitudinal survey in Spain and one of the few ones in Southern Europe, called the 'Panel of Social Inequalities in Catalonia' (PaD) (Besson et al., 2014). The PaD is the only representative household panel dataset with annual waves from 2001 to 2012 that offer key information on socioeconomic conditions, equality, justice and subjective well-being. Participation was voluntary and participants received a non-economic incentive. On average, the survey covered around 1,760 individuals per wave (PaD, 2014), and the attrition is not high from wave to wave, usually being around 8 or 9%. The final dataset for analysis comprises more than 21,000 native individual year observations, and the individuals' attrition in our final sample was 35%. Missing data were also deleted pair-wise.

Panel Analyses

Panel models allow for better control of individual unobserved fixed heterogeneities by comparing each individual with themselves. Formally, a panel model can be written as:

$$LS_{it} = \alpha + \beta x_{it} + \gamma z_i + \vartheta_{it}$$

where *LS* denotes subjective wellbeing, and is represented by life satisfaction of individual *i* in year t. The vector x_{it} refers to the number of immigrants in a particular year, indicator variables are crisis years, and a set of individual time-varying characteristics such as age, income, marital status, education, and labour market status. Time-invariant attributes such as ethnic or cultural background or migration status can only be represented by the random variable z_i if they are uncorrelated with the time-varying explanatory variables x_{it}. However, the Hausman test with both data sets demonstrates

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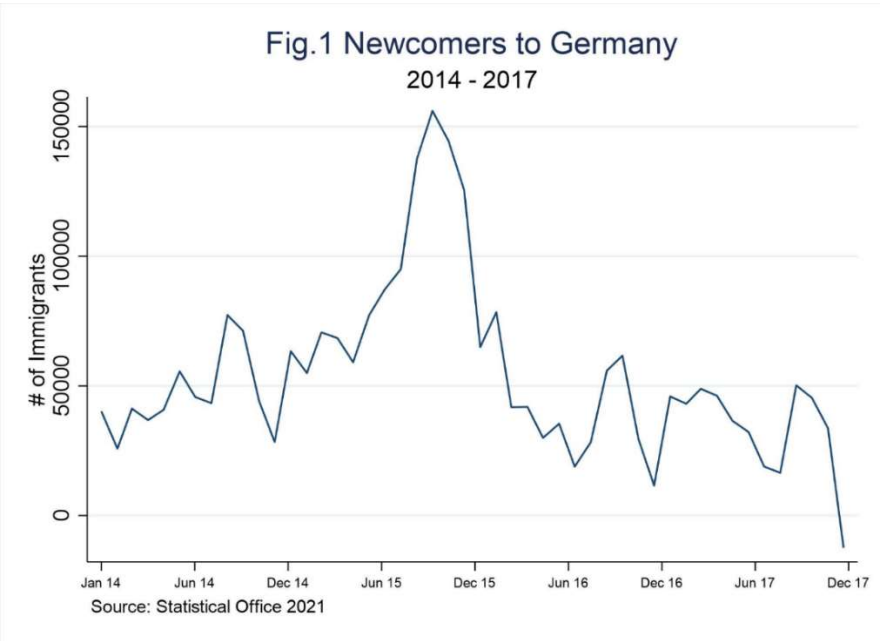
that this is unlikely. Therefore, we estimate fixed-effects models which subtract time averages and cancels the constants α and z_i from the initial models. ϑ_{it} captures idiosyncratic and normally distributed time-varying errors. To further control for auto-correlated yearly errors, we adjusted for clustered standard errors.

Regression Discontinuity Design for Germany

Disentangling correlation from causation is also a challenge in subjective well-being research (OECD, 2013) which mainly builds on observational data (Ferrer-i-Carbonell, 2013; [reference excluded to ensure anonymity]). The abrupt increase in migration numbers in 2015/16 is used to test a causal relationship between immigrants' arrival and natives' subjective well-being in Germany. The sharp increase in newcomers during the summer of 2015 resulted in the opening of the German border on 4th September 2015 and ended with an agreement between Turkey and the EU on 18th March 2016. Figure 1 illustrates this sharp increase since May 2015 and its steep drop after December 2015.



This quasi-experimental setting can be exploited using a regression discontinuity design. As GSOEP interviews start in end January and last until the end of October of each respective year. There was no systematic interview plan other than interviewee availability (Deutsches Institut für

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Wirtschaftsforschung (DIW) 2017). The random assignment to a treatment – experiencing the 'refugee crisis' in this context – is the result of the time individuals have been interviewed or the individual score value. Until May 2015, 12,931 Germans without a migration background had already answered the yearly survey, while 2,186 responded later during the rest of that year. Formally, the observed average outcome *Y* (life satisfaction) given the score *X* (time of the interview) is

$$E[Y_i|X_i] = \begin{cases} E[Y_i(0)|X_i] & if\ X_i < c \\ E[Y_i(1)|X_i] & if\ X_i \geq c, \end{cases}$$

where *c* is the cut-off (May, 2015) and *i* stands for each individual. The difference in outcomes around the cut-off is the *average* treatment effect if both regression functions $E[Y_i(1)|X_i = x]$ and $E[Y_i(0)|X_i = x]$ are continuous at *x* = *c*.

$$E[Y_i(1) - Y_i(0)|X_i = c] = \lim_{c \to x} E[Y_i|X_i = x] - \lim_{x \to c} E[Y_i\,|X_i = x]$$

Dependent and Independent Variables

Life satisfaction: The dependent variable of this study is self-reported satisfaction with life. Diener et al.'s (1985) 'Satisfaction with Life Scale' is the standard measure of subjective well-being in research on economics of happiness (e.g. Kahneman and Krueger, 2006). It is usually included in representative population surveys and asks questions such as, 'Taking all things into consideration, what is your level of satisfaction with life in general? Note that 0 is very dissatisfied and 10 is very satisfied.' This formulation was also adopted in the GSOEP and PaD panel surveys. Psychological literature points out that this is a valid measure of true inner individual happiness (Sutton and Davidson, 1997). The question in GSOEP and PaD appears in all available waves except for the PaD wave 9 (2010) and it is asked at the individual level.

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Influx of Immigrants: We used official annual and monthly migration numbers from the national statistical offices of Germany and Catalonia. Annual numbers were decomposed according to sex and for Germany also by regional background.

Socio-economic control variables: We account for time-varying individual attributes and transitions that are shown to be relevant for subjective well-being. Age, age squared (e.g. Blanchflower and Oswald, 2004; [reference excluded to ensure anonymity]), (subjective) health (e.g. Steptoe et al., 2015), net household income (e.g. Clark, et al., 2008), becoming unemployed (e.g. Dolan et al., 2008; Frey and Stutzer, 2002), and becoming married or losing a married partner (e.g. Grover and Helliwell, 2019) (see Tables 1a and 1b).

Economic crisis: Recession years define time windows in which people on average are more economically worried, miss out on economic opportunities, and face higher risks of unemployment. In Germany, the years 1993, 2002–2003, 2005, 2009, and 2012–2013 show low or negative economic growth. We contrast these years with the remaining observation years. In Catalonia, the years 2009–2012 highlight recession years.

Mass migration: Migration is a politically triggered and steered phenomenon that often changes from one year to the next (see Fig. 1). For Germany only, we identify the years 1990–1992 and 2015–2016 as time windows of mass migration which we compare to other years.

Table 1a: German Sample (1990-2017)

	N	Mean	Std. Dev.	min	max
Life Satisfaction	507188	7.033	1.782	0	10
Immigrants	523992	1004003.1	378921.56	416080	2136954
Female immigrants	523992	406334.79	133768.28	217552	770724
Male immigrants	523992	597668.3	246635.19	198528	1366230
No migration background	523992	.804	.397	0	1
1989 lived in East Germany	523993	.257	.437	0	1
Hh net income	491100	3321.729	2207.34	0	200000
Unemployed	523992	.064	.245	0	1
Poor subj. health	470180	2.612	.964	1	5
Married	523992	.592	.491	0	1
Age	523990	47.311	17.765	14	105
Tertiary education	523992	.267	.442	0	1

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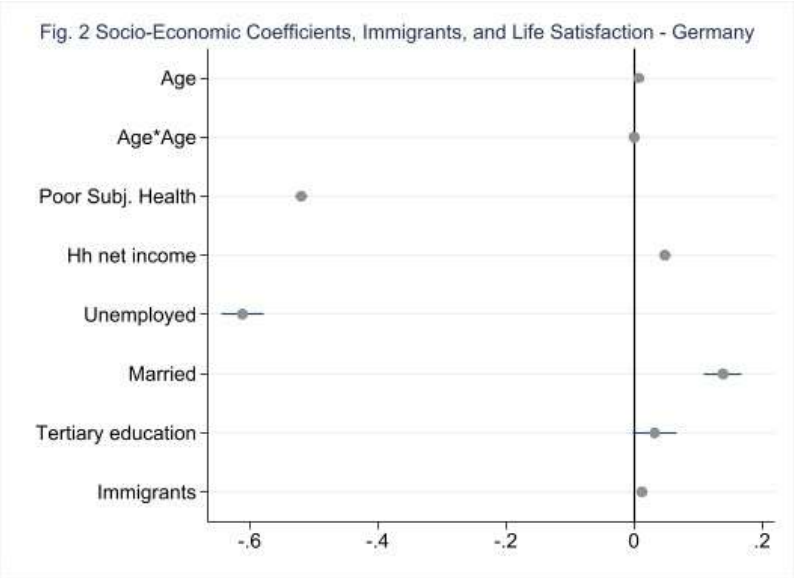
Table 1b: Catalan Sample (2001-2012)

	N	Mean	Std. Dev.	min	max
Life Satisfaction	22942	7.432	1.559	0	10
Immigrants	52709	165072.85	102284.82	55576	434499
Female immigrants	52709	55128.343	21058.227	18805	85980
Male immigrants	52709	67030.868	26740.593	24694	109683
No migration background	52709	.946	.226	0	1
HH net income	30018	1264.952	872.498	5	13165.994
Unemployed	40945	.057	.233	0	1
Poor Subj. Health	34140	2.197	.81	1	5
Married	52662	.476	.499	0	1
Age	52676	41.266	22.824	0	102
Tertiary education	44019	.222	.416	0	1

Results

Germany

Associations between socioeconomic determinants and life satisfaction are shown in Fig. 2 (see also appendix table A2). Results confirm our first hypothesis: raising numbers of immigrants (measured in 100,000 immigrants per year) significantly increase the subjective well-being of the native population (b=0.012, p <0.000). All other coefficients are also significant and shown in the original scale. To compare the relative strength of the coefficients, z-standardized variables reveal that a change in net immigration by one standard deviation leads to an increase in subjective well-being by 0.03 (p < 0.000), half the size of an increase in the net household income (β = 0.06, p < 0.000), closer to increasing the likelihood of being married (β = 0.04, p < 0.000) and a tenth of the effect of subjective health (β = 0.3, p < 0.000).



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The positive association of the influx of immigrants and the subjective well-being of the native population also holds when migration numbers are interacted with all socioeconomic determinants listed above. However, rising numbers of newcomers do not significantly interact with being unemployed, married, unhealthy, or older. Only higher net household incomes (in 1,000€) interact negatively with higher migration numbers ($b = -0.002$, $p < 0.000$). Moreover, after controlling for these conditions, higher immigration numbers are more positively associated with the subjective well-being of East-Germans than West-Germans. We keep all significant interactions in the following models.

Table 2 - Immigration and Life Satisfaction of Natives - Germany

	(3) FE	(4) FE
Immigrants	.017***	.014***
	(.001)	(.002)
Economic Crisis years	−.04*	−.075***
	(.021)	(.022)
Economic Crisis years × Immigrants	.003	.007***
	(.002)	(.002)
Mass migration years		−.212***
		(.059)
Mass migration years × Immigrants		.013***
		(.003)
Observations	356597	356597
R^2	.088	.088
Socio-economic controls	YES	YES

Robust standard errors are in parentheses
**** p < .01, ** p < .05, * p < .1*

Models 3 and 4 in Table 2 confirm that years of economic crisis and years of mass migration are disruptive experiences of insecurity which significantly lower subjective well-being of people (*Hypothesis 2*). But they positively interact with numbers of immigrants. Moreover, in these models, immigration numbers still preserve their main positive associations with life satisfaction ($b = 0.017$, $p < 0.001$; $b = 0.014$, $p < 0.002$). All associations are also robust with lagged covariates, expect for the vanishing effect of mass migration years and their negative interaction with immigrants. As (German) migration policies often abruptly close borders, this change is not surprising (not shown).

More importantly, if we further decompose the population of immigrants by gender, we see that an increase in life satisfaction is only associated with higher numbers of female migrants. In

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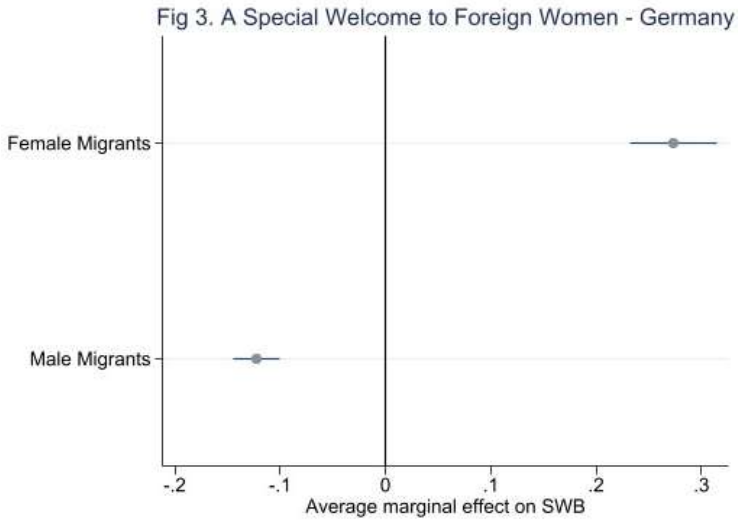
contrast, the arrival of male newcomers is negatively related with natives' subjective well-being. The difference is substantial. Fig. 3 displays the marginal effects of the nested model 5 which supports Hypothesis 3 (see appendix Table A2).



This gender effect also holds when immigrants are separated into regions of origin. Females from all corners of the world are positively correlated with the subjective well-being of the local population in Germany. The only exemptions are women from Australia, where we see no association and women with no citizenship where the coefficient is negative. Given the small sample size of the latter subgroup, the confidence intervals are very wide represented by the strokes around the point estimates. In contrast, male immigrants from various world regions are predominantly negatively associated with natives' subjective well-being (Fig. 4).

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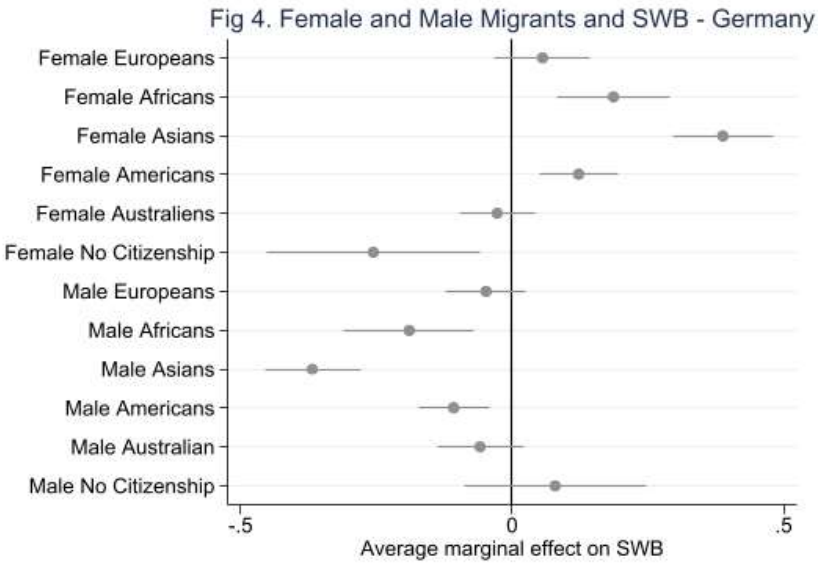

Catalonia, Spain

We replicated the analysis using Spanish (Catalan) panel data. Figure 5 below shows the same significant associations of socio-economic determinants and life satisfaction as in Germany. The only exception is unemployment. The positive association is determined by the recession years after 2008.[1] During times of extreme unemployment, the individual destiny of losing one's job is less detrimentally experienced. Coefficients are shown in the original scale. Immigration is measured per 1,000 migrants. On this scale, positive immigration numbers do significantly increase the subjective well-being of the native population by (b = 0.003, p < 0.000) and confirm our first hypothesis. A change in net immigration by one standard deviation leads to an increase in subjective well-being by 0.356 (p < 0.000), which is somehow similar to effect of subjective health (β = 0.22, p < 0.000) and relatively more than the size of an increase in educational level (β = 0.15, p < 0.010).

[1] Spain suffered the highest growth of unemployment and inequality in the EU as a result of the 2008 Economic Crisis (OECD 2016). From 2007 to 2012, the unemployment rate rose from 8.57% to 25.77%, particularly affecting young people (from 18.1% for those aged 16 to 24 years old in 2007 to 52.9% in 2012) (Spanish Statistical Office 2021b). Catalonia closely followed the average Spanish unemployment rates and GINI index before and after the 2008 Economic Crisis (Instituto Nacional de Estadistica (INE) 2022).

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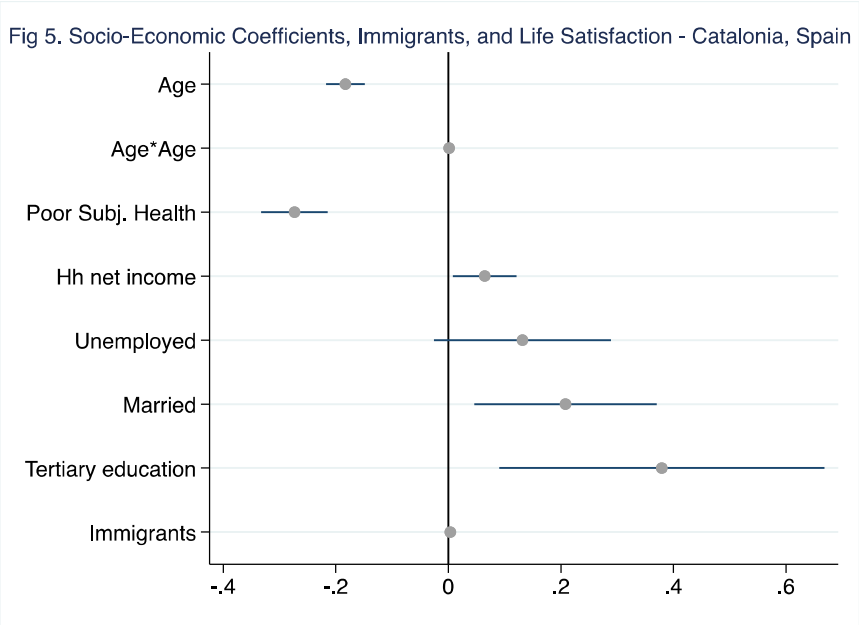

The overall positive association of immigration and subjective well-being shifts when we replicate the previous model 4 with Catalan data. Model 7 in Table 3 does this and certifies that the economic crisis years after 2008 significantly lowered the subjective well-being of Catalan people (*Hypothesis 2*). However, the interaction between crisis years and objective immigration numbers positively deviates from the negative main crisis effect and also neutralizes the negative immigration effect for these years. Moreover, while being unemployed interacts positively with the 2008 Economic Crisis, the main effect of being unemployed is negative (see appendix Table A3).

Table 3 - Immigration and Life Satisfaction of Natives – Catalonia, Spain

	(6) FE	(7) FE
Immigrants	.003***	-.001***
	(0)	(0)
2008 Economic Crisis		-4.365***
		(.251)
2008 Economic Crisis × Immigrants		.033***
		(.002)
Observations	14659	14659
R^2	.040	.081
Socio-economic controls	YES	YES

Robust standard errors are in parentheses
**** p<.01, ** p<.05, * p<.1*

More importantly, if we decompose the population of immigrants, we see that the increase in life satisfaction relates only to a higher influx of female immigrants. In contrast, male newcomers were negatively associated with subjective well-being in Catalonia. Figure 6 displays the marginal effects of

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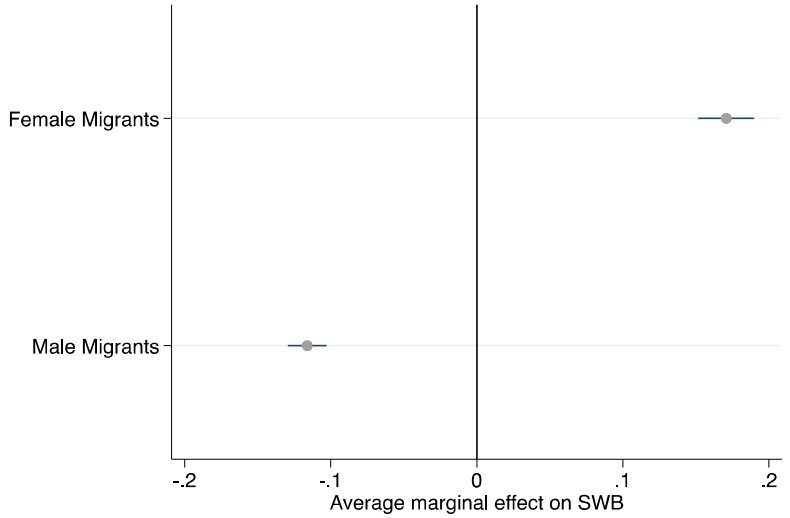
the nested Model 8 (see appendix Table A3). This is in line with our German results and thus confirms our third hypothesis.

Fig 6. A Special Welcome to Foreign Women - Catalonia, Spain



Given the small dataset, we could not fully replicate the model shown in Fig. 4. Nonetheless, we tested our hypothesis with different subgroups of immigrants (i.e. European, American, African, stateless). The results still show that while immigrant women have a positive relationship with the subjective well-being of natives, the relationship turns negative for immigrant men.

Causal Evidence? Regression Discontinuity Analysis for Germany

To further test if the influx of newcomers causes changes in the subjective well-being of the native population, we zoom into the years 2014–2016 when Germany first experienced a steep increase and then a sharp decline in immigration numbers mainly from Syria, Iraq and Afghanistan. The so-called "migration crisis", as the high inflow of immigration in 2015 was often described, may be perceived as a natural field experiment treatment.[2] The numbers sharply increased in January and June and peaked in September 2015. We took the date of the interview as a random event and assigned

[2] Natural field experiments are conducted outside a lab and in a way that participants are, a priori, unaware of participating in a scientific study (Harrison and List 2004). This is important because it avoids socially desirable answers and self-selection of participants.

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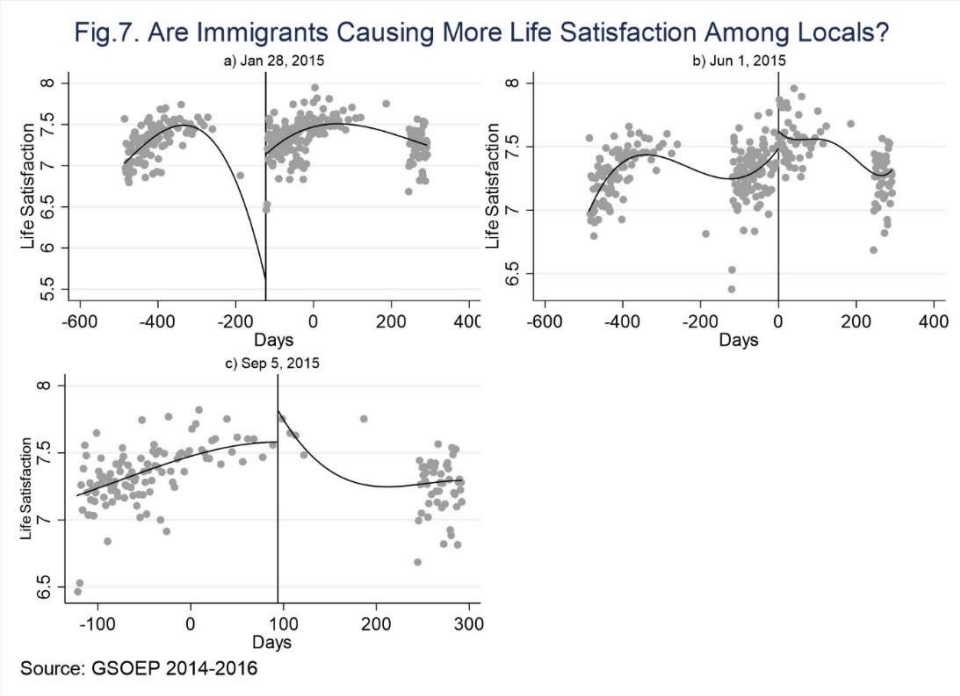
interviewees into a control group who were interviewed before the respective cut-off time and a treatment group who were interviewed thereafter.



The Regression Discontinuity plots in figure 7 show the change in life satisfaction at the time when newcomers arrived in large numbers. The first plot, for 28 January, displays the earliest measured response to the increasing number of immigrants in the GSOEP 2015 data. However, the sharp difference before and after the cut-off may also be due to the fewer observations that were collected during autumn 2014. Still, the pattern is replicated both in June and September when the German chancellor decided to ignore the Dublin regulation and took in thousands of immigrants. Our observational window was closed by the Turkey Agreement on 18 March 2016. After that, immigration stopped abruptly. Robust point estimation for June confirms a robust positive effect of 0.28 (z = 83.8, p < 0.000) at the cut-off. The robust effect for 5 September is 0.199 (z = 3.356, p < 0.001).

Discussion and Conclusions

The literature on migration and subjective well-being largely neglects the life satisfaction and happiness of the native population in host countries (e.g. Banerjee et al., 2019; Sand and Gruber, 2018). Sparse empirical research provides contradictory findings. We hypothesized first that rational

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actors should experience more subjective well-being when more migrants enter the country as they add human capital and other material and immaterial resources to the national economy and community. But second, we expected that abrupt changes like sudden influxes of migrants or economic crises may evoke ethnic competition and may decrease the subjective well-being of the native population. Finally, and more innovative, we assumed that gender norms shape attitudes towards female and male migrants and alter the subjective well-being of the host population.

Our results show that immigration positively correlates with the subjective well-being of the native population, irrespective of the national or ethnic composition of immigrants. This positive relationship reflects the overall increase in human resources and productivity. Also, we see that this relationship holds even during times of economic crisis and of abruptly increasing migration numbers in Germany. To be sure, crisis years with rising social insecurity are associated with lower levels of life satisfaction in general. Yet, the interaction between the crisis years and the number of immigrants is significantly positive in Germany and Catalonia, controlling for other socio-economic determinants. Other researchers have identified this positive nexus between rising numbers of newcomers and subjective well-being on the level of growing and shrinking cities (Glaeser et al., 2016). Conversely, this finding means immigration opposition and xenophobia can be counteracted by socio-economic assimilation. However, it is beyond the scope of this paper to identify solid mechanisms.

Perhaps even more intriguing, we find that the sex composition of immigrants plays a crucial role in the subjective well-being of natives. If we divide the influx of immigrants by gender and by their continent of origin, we clearly see that while female immigrants are positively associated with the life satisfaction of the native population, male migrants produce negative correlations. This gender difference holds for both country cases and for migrants from all world regions and can help to understand the conflicting findings of the literature.

Stereotypical gender expectations may explain the positive attitude to female immigrants. All over the world, research shows that women (still) spend more time on house and care work than men (Treas and Drobnič, 2020). What is more, better educated and emancipated women also outsource

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distressing house and care work mostly to women with lesser education and lower hourly earnings such as nannies, cleaning ladies, kindergarten teachers, dog walkers, and elderly care nurses. These jobs are often taken up by female immigrants (Chammartin, 2006). One can easily imagine how full-time employed women and men in host countries embrace this support. In addition, women are expected to be less competitive, more compliant, and less violent than men, which is why male immigrants are more often perceived as a threat to peace, stability, and happiness. Immigrant women may also increase the supply of potential mating partners and thereby decreasing male competition. Finally, a biased (social) media coverage of negative and deviant events as well as a xenophobic political discourse may contribute to the negative relationship between the influx of male migrants and the subjective well-being of natives (Benesch et al., 2019; Eberl et al., 2018)

It is beyond the scope of this analysis to disentangle the importance and the dynamics of (social) media and people's anti-immigration attitudes on their subjective well-being. There are also other limitations to this study that should be addressed in future research. The positive association of the number of migrants and the subjective well-being of natives is operationalized as a linear effect. This is the first overall approximation, but subsequent studies should test for non-linearity and tipping points to provide a fuller picture. Along similar lines, the impact of times of crisis needs to be further decomposed, particularly for different socioeconomic groups. Further, we tried to make our causal presumptions safer by using different designs and panel and regression discontinuity models. Nonetheless, even our rich panel datasets and ambitious model strategy do neither allow to fully proof our causal claims nor to fully replicate all aspects of the analysis for the Catalonian case and to isolate the gender effect in the regression discontinuity analysis. This study should be cross-validated using further datasets and experimental designs. Particularly, future research should explore how well findings from Western countries travel to countries with more extreme collectivistic values and other levels of economic development such as those in Southeast Asia, Islamic or Orthodox regions or countries from the Global South (see: Diener and Suh, 2000; Hofstede et al., 2005).

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Overall, in terms of policy implications, our findings could potentially have an important positive influence on future migration and social cohesion policies. In particular, given the lack of a gender perspective in migration, asylum and integration policies (European Institute for Gender Equality, 2020), the sex-composition effect should shape lawmakers' approach to migration policies in the future. Our findings suggest that future policies need to better balance the sex ratio of immigrants, for example, by enrolling migration programs that particularly invite women to come and stay. These policies would increase the well-being of the native population, and since happiness is contagious, it will also improve the well-being of immigrants. Moreover, women-focused migration policies may likely radiate to the sending countries, possibly raising the status of women there and triggering other positive changes that come with gender equality.

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Appendix

Table A2 - Immigration and Life Satisfaction of Natives – Germany

	(1) FE	(2) FE	(3) FE	(4) FE	(5) FE
Age	.007***	.008***	.008***	.007***	.003
	(.002)	(.002)	(.002)	(.002)	(.003)
Age × Age	0***	0***	0***	0***	0***
	(0)	(0)	(0)	(0)	(0)
Poor Subjective Health	-.52***	-.523***	-.52***	-.52***	-.519***
	(.005)	(.01)	(.005)	(.005)	(.005)
Immigrants	.012***	.018***	.017***	.014***	
	(.001)	(.003)	(.001)	(.002)	
Female Immigrants					.402***
					(.036)
Male Immigrants					-.188***
					(.018)
Hh net income (1,000€)	.049***	.063***	.063***	.063***	.083***
	(.004)	(.004)	(.004)	(.004)	(.006)
Hh net income × Female Immigrants					-.041***
					(.006)
Hh net income × Male Immigrants					-.188***
					(.018)
Economic Crisis years × Hh net income			-.002***	-.002***	
			(0)	(0)	
Unemployed	-.611***	-.667***	-.611***	-.611***	-.608***
	(.017)	(.039)	(.017)	(.017)	(.017)
Married	.139***	.13***	.139***	.14***	.142***
	(.015)	(.021)	(.015)	(.015)	(.015)
Tertiary education	.033*	.041*	.033*	.032*	.029
	(.018)	(.022)	(.018)	(.018)	(.018)
Immigrants × E-Germany		.003*			
		(.002)			
Immigrants × Poor Health		0			
		(.001)			
Immigrants × Unemployed		.006			
		(.004)			
Immigrants × Hh net income		-.001***			
		(0)			
Immigrants × Married		.001			
		(.001)			
Immigrants × Age		0			
		(0)			
Immigrants × Tert. Education		(.001)			
Economic Crisis years			-.041*	-.075***	-.108***
			(.021)	(.022)	(.033)
Economic Crisis years × Immigrants			.003	.007***	
			(.002)	(.002)	
Economic Crisis years × Female Immigrants					.054
					(.066)
Economic Crisis years × Male Immigrants					-.017
					(.038)
Mass migration years				-.212***	-.302***
				(.059)	(.052)
Mass migration years × Immigrants				.013***	
				(.003)	
Mass migration years					-.065**

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× Female Immigrants					(.025)
Mass migration years × Male Immigrants					.073*** (.014)
Constant	7.964***	7.898***	7.915***	7.983***	8.005***
	(.066)	(.068)	(.066)	(.068)	(.073)
Observations	356597	356597	356597	356597	356597
R^2	.087	.088	.088	.088	.089

Robust standard errors are in parentheses
**** p<.01, ** p<.05, * p<.1*

Table A3 - Immigration and Life Satisfaction of Natives – Catalonia, Spain

	(6) FE	(7) FE	(8) FE
Age	-.183***	-.0459**	-.350***
	(.0175)	(.020)	(.023)
Age × Age	.001***	.001***	.001***
	(0)	(0)	(0)
Poor Subjective Health	-.273***	-.273***	-.273***
	(.030)	(.030)	(.030)
Immigrants	.003***	-.001***	
	(0)	(0)	
Female Immigrants			172***
			(.009)
Male Immigrants			-.117***
			(.006)
Hh net income (1,000€)	.0647**	.008	.00570
	(.028)	(0.02	(.0275)
Unemployed	.132	-.128	.128
	(.080)	(.121)	(.0822)
2008 Economic Crisis × Unemployed		.352** (145)	
Married	.208**	.185**	.183**
	(.082)	(.081)	(.0817)
Tertiary education	.355**	.363**	.358**
	(.147)	(.143)	(.143)
2008 Economic Crisis		-4.365***	.899***
		(.251)	(.0809)
2008 Economic × Immigrants		.033*** (.002)	
Constant	12.75***	7.029***	19.39***
	(.421)	(.582)	(0.745)
Observations	14659	14659	14659
R^2	.040	0.081	.081

Robust standard errors are in parentheses
**** p<.01, ** p<.05, * p<.1*

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